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Title: Cybersecurity in the Age of Generative AI: A Systematic Taxonomy of AI-Powered Vulnerability Assessment and Risk Management

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Summary:

This paper provides a systematic review of the role of Generative AI (GenAI) in transforming vulnerability assessment (VA) and risk management (RM) within cybersecurity, covering the entire lifecycle from vulnerability discovery to remediation. Through an analysis of over 100 publications from 2021 to 2025, the authors develop a comprehensive taxonomy that categorizes GenAI's applications into offensive (e.g., simulating advanced threats like APTs or generating exploits) and defensive (e.g., automating detection, patching, and hardening) strategies. The taxonomy highlights dominant GenAI techniques, such as large language models (LLMs) for threat intelligence, generative adversarial networks (GANs) for anomaly detection, and integration with Security Orchestration, Automation, and Response (SOAR) systems to accelerate response times. Key contributions include mapping GenAI's impact on addressing traditional challenges like scalability in IoT environments and evolving cyber threats, emphasizing its potential to enhance resilience through automated, proactive measures.

The methodology involves a structured literature review, employing keyword searches across databases like Scopus, IEEE Xplore, and ACM Digital Library, with inclusion criteria focused on peer-reviewed works on GenAI in VA/RM. Findings reveal GenAI's strengths in automating labor-intensive tasks, such as code vulnerability scanning and risk prioritization, but also underscore challenges including security vulnerabilities in GenAI itself (e.g., prompt injection attacks), lack of explainability leading to "black box" decisions, and trustworthiness issues like biases or hallucinations in outputs. The authors advocate for Explainable AI (XAI) to enable transparent, auditable, and human-validated risk assessments, ensuring ethical deployment.

In conclusion, the paper posits that GenAI offers a paradigm shift in cybersecurity by resolving longstanding VA/RM limitations, but calls for interdisciplinary research to tackle its risks. It positions GenAI as a fertile area for innovation, recommending frameworks for secure integration and policy guidelines to balance efficiency with accountability. While no specific tables or figures are detailed in accessible content, the taxonomy likely serves as a central framework, potentially visualized in diagrams outlining offensive/defensive categories and lifecycle stages. This work serves as a foundational reference for researchers and practitioners aiming to leverage GenAI in building more robust cybersecurity systems.